GCP Architecture for the Customer Churn Feature Mart

Proposed design | Grain: one customer per calendar month

PROPOSED INGESTION - source formats and compatibility require confirmation

Profiles, contracts, support, orders

Change capture: profiles and contracts daily; support and orders hourly.

Invoices and usage

Append feeds: daily invoices; hourly usage micro-batches.

Inventory and geography

Full snapshots: daily inventory; monthly geography.

Change capture

Datastream if compatible; otherwise source-specific extraction.

File exports

Design prerequisite: sources can produce files suitable for BigQuery load jobs.

Cloud Storage

Dated raw deliveries; retained batches for replay and recovery.

BIGQUERY - the assessment assumes raw tables already exist

BigQuery load jobs and raw tables

Load batches at source delivery frequencies; track replayed batches.
Required history: retained versions with _ingested_at; complete snapshots with _snapshot_date.

SQL preparation in BigQuery

Temporary tables: historical version availability intervals and selection of the relevant snapshots.

SQL feature calculations and assembly

Aggregate to customer-month grain before the final joins.
Profile, contracts, usage, payments, support, orders and geography.

Pre-publication checks

Check version conflicts upstream; validate keys, uniqueness and row counts on the candidate result.

Pass

Transactional publication

Replace requested months: DELETE followed by INSERT.
COMMIT validates both operations.
Outside session mode, an unhandled script error rolls back the open transaction.

mart_churn_features_flat

One row per customer_id and billing_month.
PARTITION BY billing_month ; CLUSTER BY customer_id.

Managed Airflow on Google Cloud

Scheduling, dependencies, targeted retries and historical backfills.
Verify expected deliveries; pass sources_ready to the SQL script.
Prevent concurrent publications with overlapping month ranges.

Failed checks

Block publication and preserve existing data. Airflow reports the failure and coordinates retries.

Proposed optimization

Usage: partition by consumption date USAGE_DT.
Filter dates and required columns.
Select applicable versions before monthly joins.
Raw partitioning is recommended, not confirmed by the supplied schema.

Data Science team

Query features directly.

Solid arrows: data flow

Dashed arrows: Airflow control

sys_etl_job_logs: operational monitoring; excluded from customer features.